

Blood Cell Image Classification with K-Nearest Neighbors (KNN)

Supervised

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Outline

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Problem and Goal

- Input: microscopic blood cell image (JPEG)
- Output: one of 4 classes (EOSINOPHIL, LYMPHOCYTE, MONOCYTE, NEUTROPHIL)
- Goal: build an end-to-end classical image classification pipeline and evaluate it rigorously
- Constraint: use “basic” ML (no deep CNNs); compare feature representations

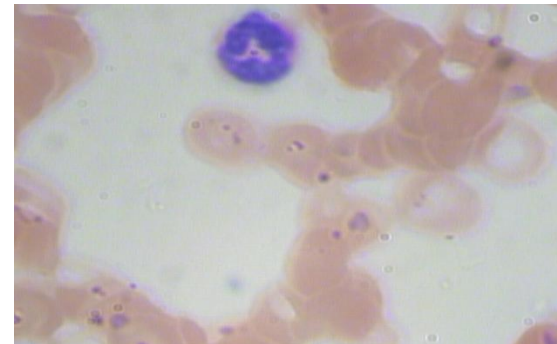
DATASET DESCRIPTION

The Blood Cell Classification Dataset from Kaggle [1]. It contains roughly 12,500 augmented images.

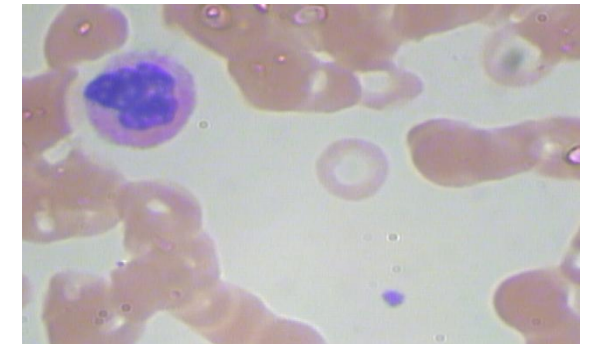
Label	No of Images
Neutrophil	3123
Eosinophil	3120
Lymphocyte	3103
Monocyte	3098
Total	12444

Image Size: 240 * 320

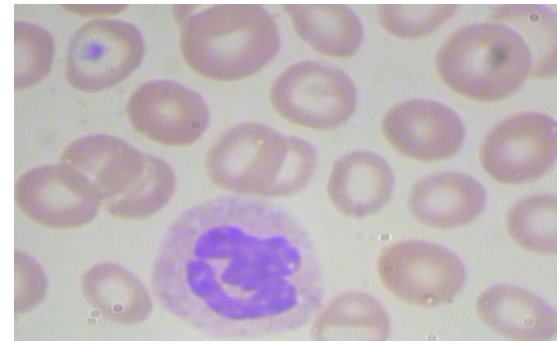
Eosinophil



Lymphocyte



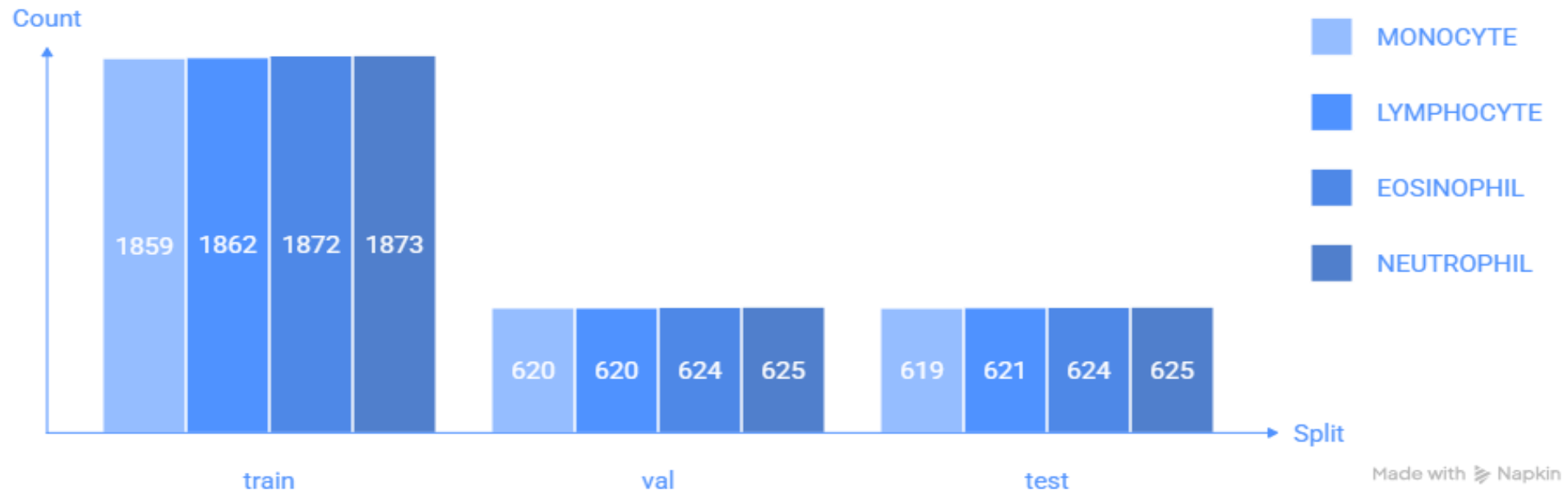
Monocyte



Neutrophil



Train / Validation / Test split

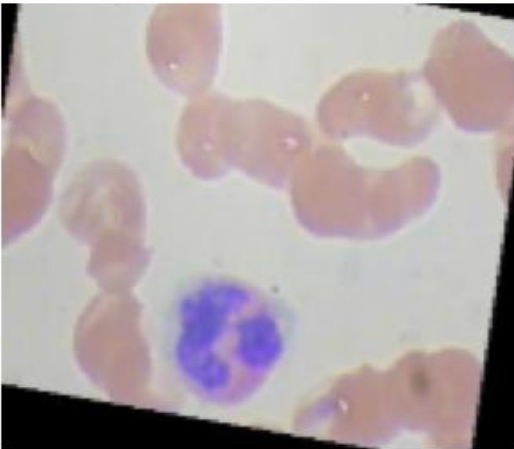


Preprocessing

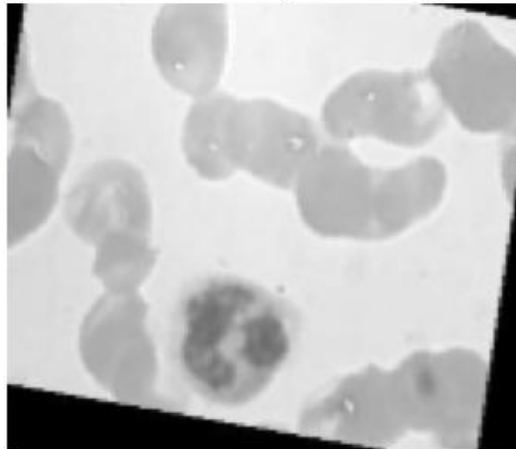
1. Resize: consistent feature dimensions (128 * 128)
2. Grayscale: simplifies edges/texture/frequency features
3. Denoising: Gaussian blur just to reduce noise
4. CLAHE: improves local contrast in microscope images

Preprocessing Pipeline Stages on Sample Image 1

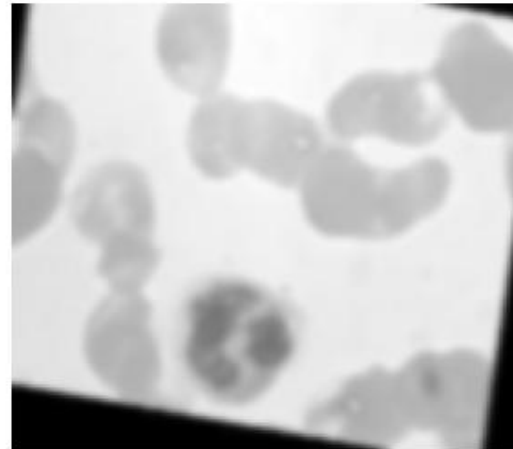
Resized



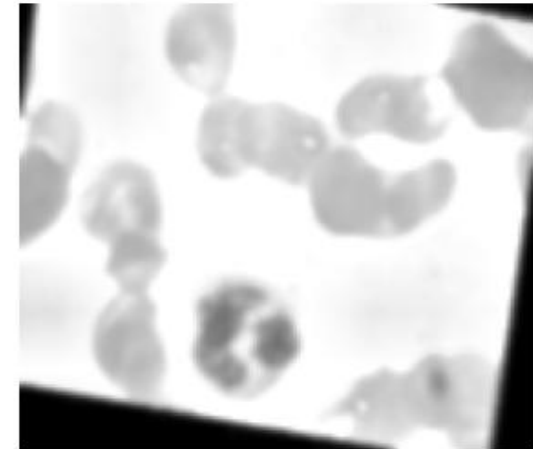
Gray



Gaussian Blur

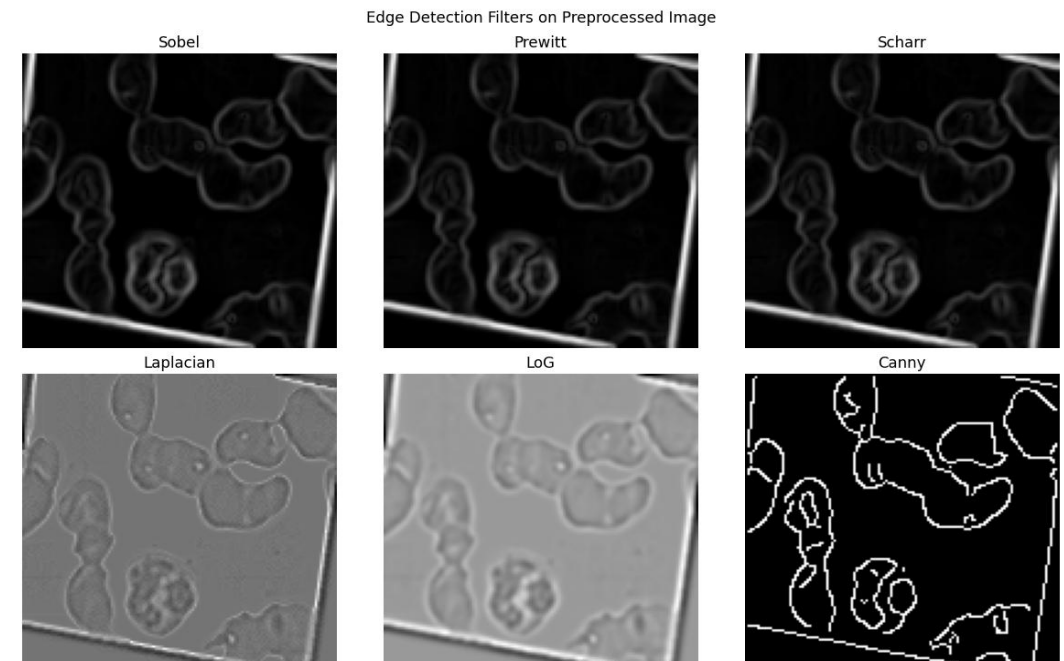
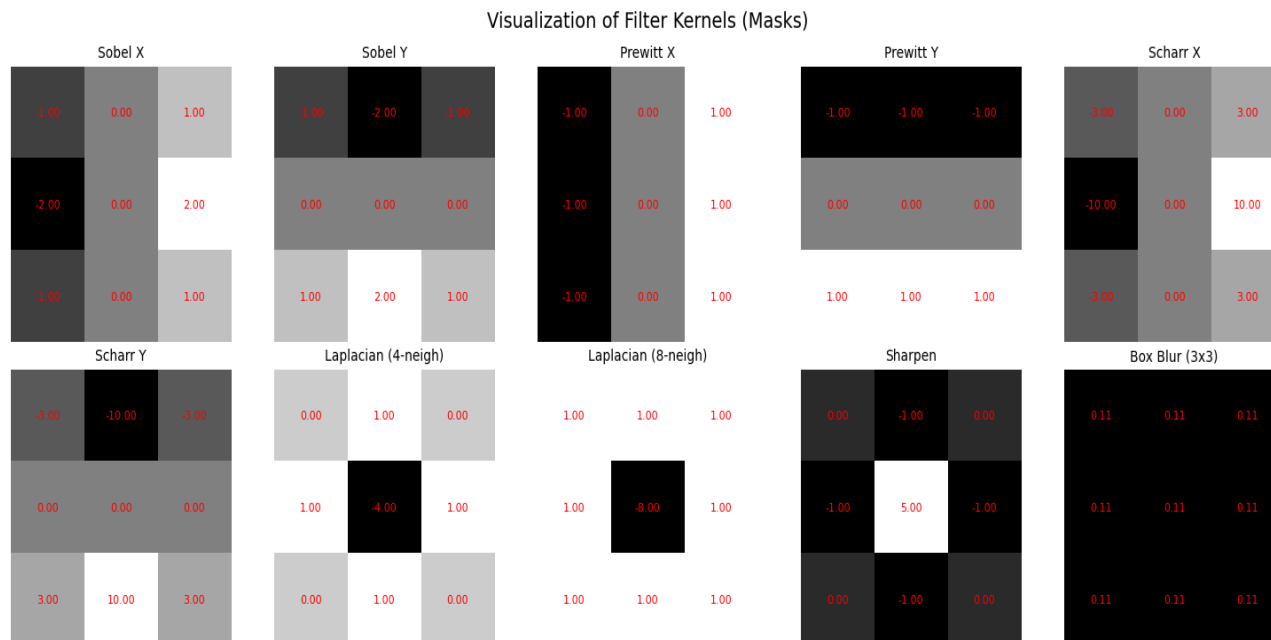


CLAHE



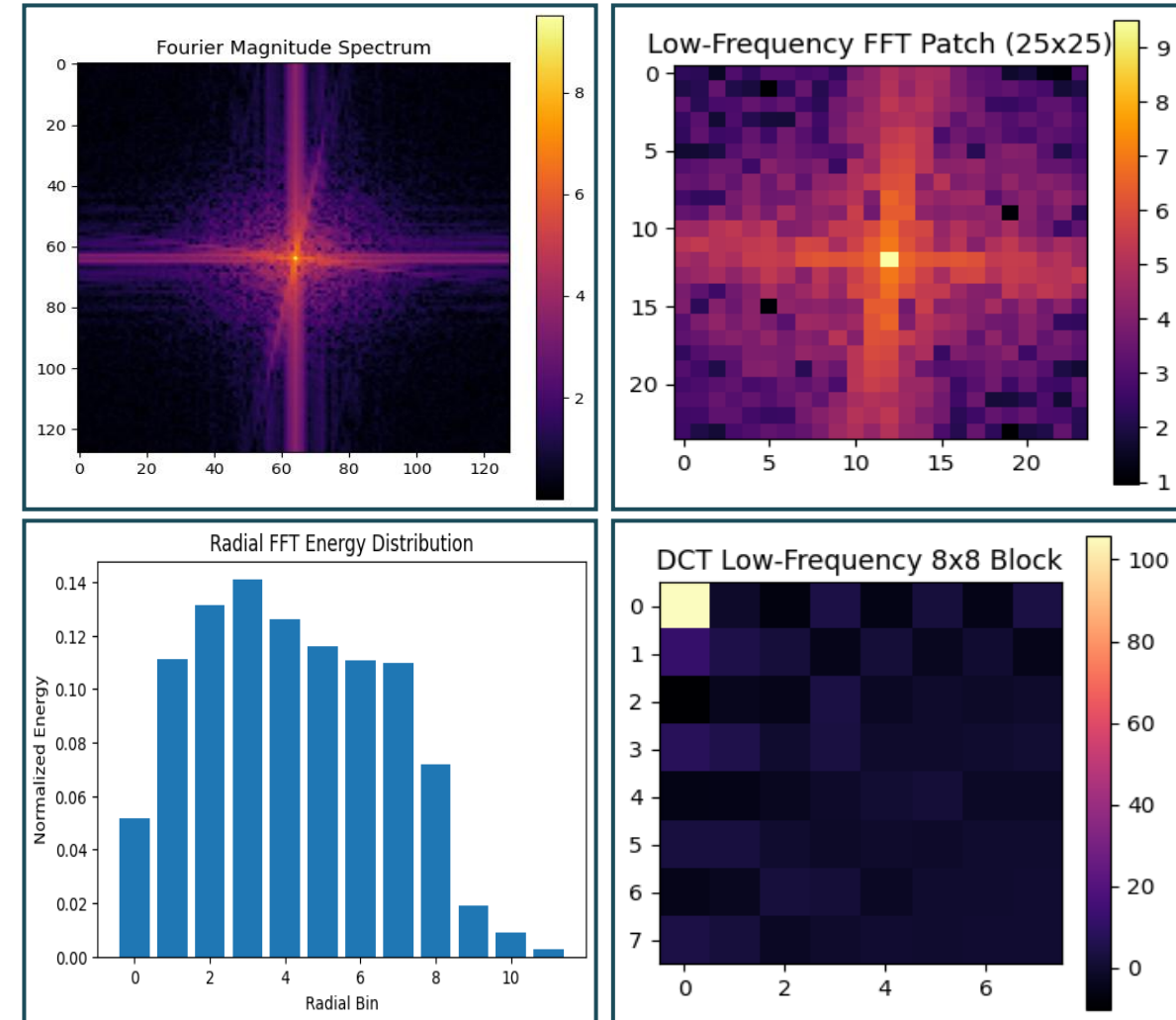
Filters

- Edges: Sobel / Laplacian / LoG / Canny can capture boundaries
- Texture: LBP, GLCM (Haralick properties) can captures micro-patterns
- Frequency: FFT magnitude + radial energy; DCT low-freq block can captures global structure
- Morphology: Otsu threshold + opening/closing can shape descriptors (area, perimeter, roundness)
- These are not just visuals, they feed into feature vectors for KNN.



Frequency-Domain Image Analysis

- Fourier magnitude spectrum shows how image energy is distributed across spatial frequencies; strong center indicates dominant low-frequency (global) structure.
- Low-frequency FFT patch & DCT 8×8 block compactly capture coarse shape and smooth intensity variations.
- Radial FFT energy distribution summarizes frequency content independent of orientation, distinguishing smooth vs textured images.



Feature pipelines

1. KNN relies on distances , raw pixels often fail because of lighting/pose variation
2. So, I convert each image into a feature vector that is more stable and discriminative
3. Each pipeline = (preprocessing choices) + (feature types) → then KNN classification
4. I've compared pipelines fairly using the same split and the same metric reporting

Feature pipelines

Pipeline Name	Feature Names	Feature Size (per image)	Why
RawPixels	raw_pixels	16,384	Baseline(grayscale flatten)
GrayHist	gray_hist	32	global brightness
ColorHists	rgb_hist, hsv_hist	96	stain/color differences
Texture_LBP_GLCM	lbp_hist, glcm_texture	67	micro texture patterns
EdgesAndTexture	sobel_hist, lbp_hist	51	Shape and texture cues
Frequency_FFT_DCT	fft_radial, dct_low_block	72	global frequency/shape
Morphology_GrayHist	shape_morphology, gray_hist	38	size/shape cues
Combo_BestGuess	gray_hist, lbp_hist, sobel_hist, fft_radial	91	strongest combined representation

KNN methodology

- Training: store feature vectors + labels
- Prediction: compute distances to all training points , take k nearest. majority vote
- Tune k on the validation set and then evaluate the chosen configuration on the test set
- K controls bias-variance, small k overfits, large k smooths.
- **L2 distance** : is the **Euclidean distance** between two feature vectors.
- K = 3, 4, 5

$x \rightarrow$ feature vector of one training image

$y \rightarrow$ feature vector of one validation/test image

$n \rightarrow$ feature dimension of the pipeline e.g. 32 (GrayHist)

$$d(x, y) = \sqrt{\sum_{i=1}^n (y_i - x_i)^2}$$

Evaluations

- Accuracy + Macro Precision/Recall/F1 + Confusion Matrix

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FP + TN + FN)}$$

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

$$\text{Micro - Precision} = \frac{\text{TruePositives1} + \text{TruePositives2}}{\text{TruePositives1} + \text{FalsePositives1} + \text{TruePositives2} + \text{FalsePositives2}}$$

$$\text{Micro - Recall} = \frac{\text{TruePositives1} + \text{TruePositives2}}{\text{TruePositives1} + \text{FalseNegatives1} + \text{TruePositives2} + \text{FalseNegatives2}}$$

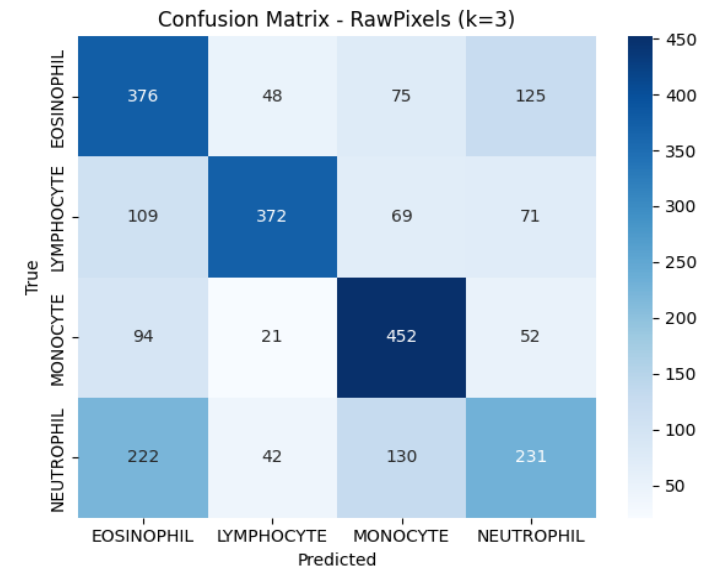
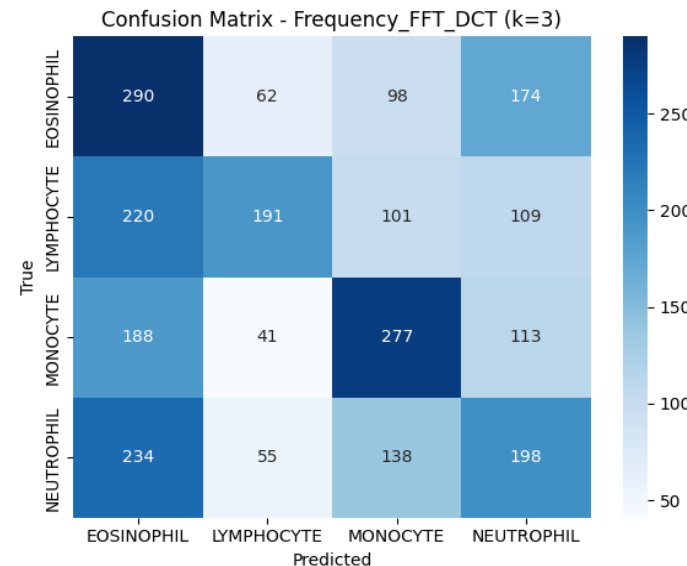
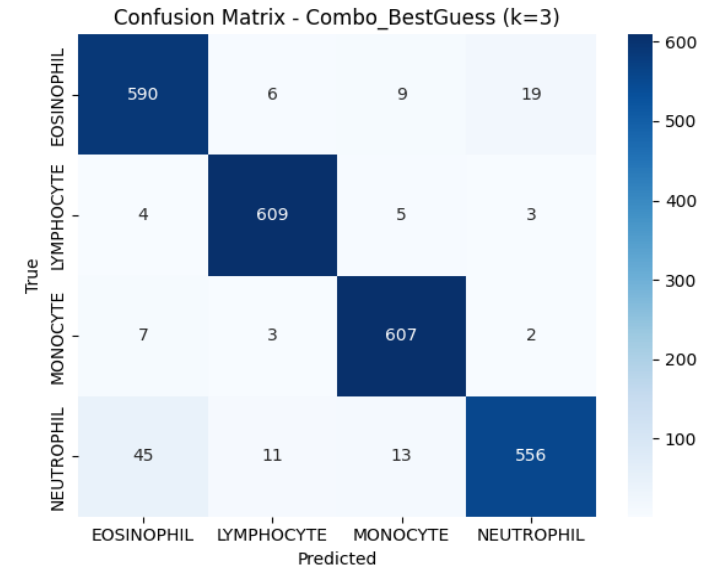
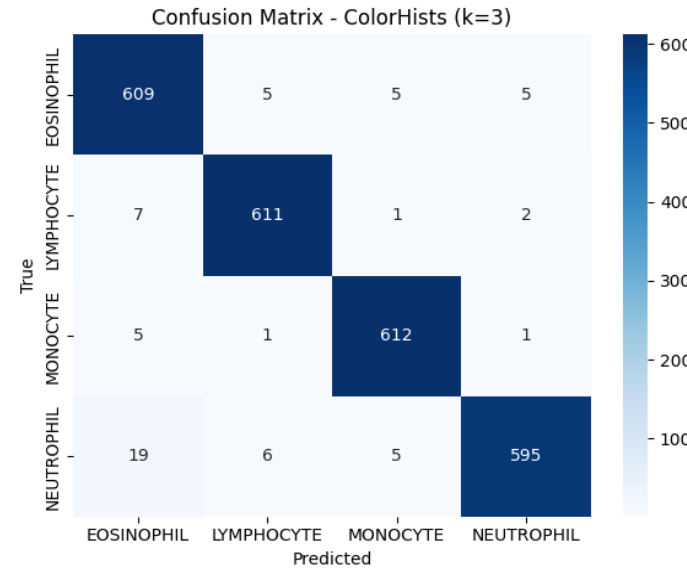
$$\text{Micro - F - Score} = 2 \cdot \frac{\text{Micro - Precision} \cdot \text{Micro - Recall}}{\text{Micro - Precision} + \text{Micro - Recall}}$$

Results

Rank (by Macro F1)	Pipeline	Accuracy	Macro Precision	Macro Recall	Macro F1	k
1	ColorHists	0.960627	0.961172	0.960731	0.960625	3
2	Combo_BestGuess	0.928887	0.930937	0.929091	0.928289	3
3	GrayHist	0.894737	0.899283	0.895020	0.893401	3
4	Morphology_GrayHist	0.856167	0.863453	0.856518	0.854216	3
5	EdgesAndTexture	0.799518	0.800708	0.799984	0.795155	3
6	Texture_LBP_GLCM	0.766171	0.768577	0.766687	0.759488	3
7	RawPixels	0.498192	0.515854	0.498469	0.495947	3
8	Frequency_FFT_DCT	0.353154	0.373577	0.353135	0.352148	3

Confusion Matrix

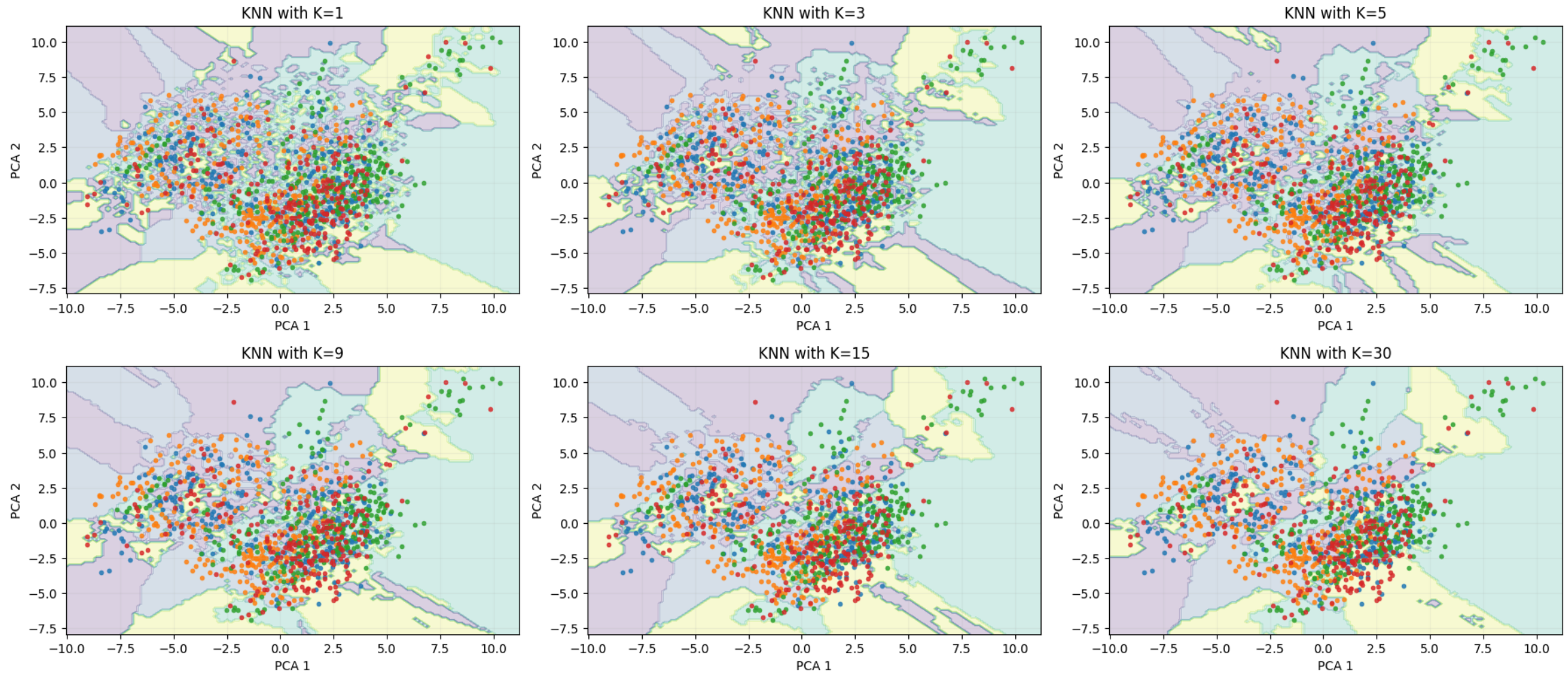
- Microscope staining/lighting creates strong color cues between WBC subtypes
- Color histograms are simple but robust to small translations/rotations
- K=3 performed best, nearest-neighbor in color space is already very discriminative
- Raw pixels performed poorly, sensitive to background, positioning, illumination
- Frequency_FFT_DCT performed poorly, global frequency features did not separate these classes well



KNN behavior vs k (visualization in PCA space)

KNN Decision Boundaries (PCA 2D) | Pipeline = ColorHists

0 - EOSINOPHIL 1 - LYMPHOCYTE 2 - MONOCYTE 3 - NEUTROPHIL



Conclusion

- Best approach: ColorHists + KNN (k=3)
- Validation macro-F1 ≈ 0.964 ; Test accuracy ≈ 0.970
- Feature choice mattered much more than the classifier (raw pixels performed poorly)
- Saved splits + feature matrices + evaluation outputs (The pipeline is reproducible)
- Simple, well-chosen features + KNN can reach very high performance on this dataset

Future Work

- Robustness & sensitivity like noise, brightness, reduced training size
- Use bounding boxes to crop cells more precisely before feature extraction
- Compare against Naive Bayes, and k-means as an unsupervised baseline
- Compare with a small CNN baseline to contextualize results

References

[1] <https://www.kaggle.com/datasets/paultimothymooney/blood-cells>

Thank you